

SekolahPro

System Architecture & Flow Overview

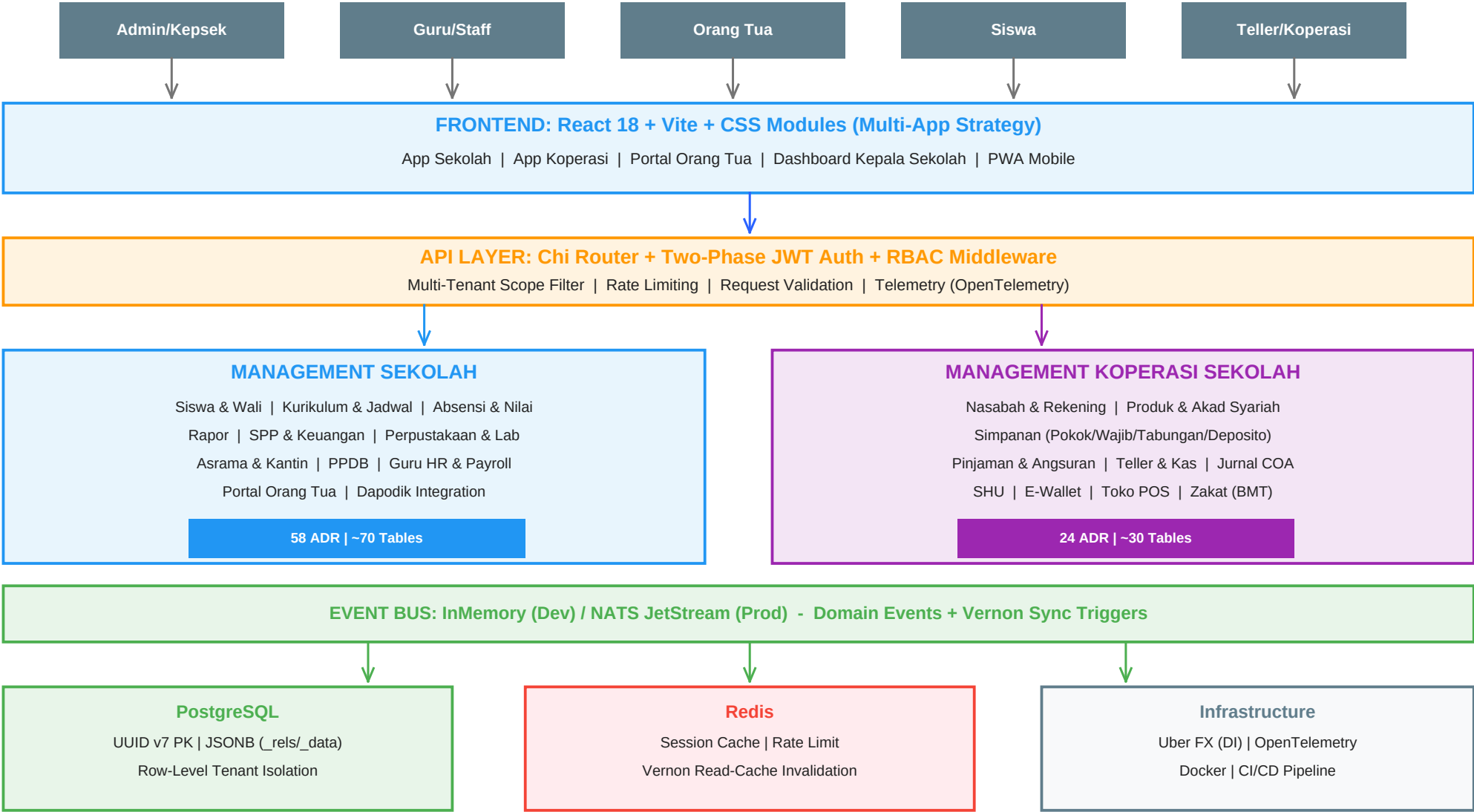
Management Sekolah + Management Koperasi Sekolah
Dual-Mode: General (Umum) & Islamic (Pesantren/BMT)

Tech Stack: Go (Clean Architecture + CQRS) | React 18 + Vite
Vernon Denormalized Read-Cache | Multi-Tenant 4-Level Hierarchy
Event-Driven (NATS JetStream) | PostgreSQL + Redis

100 ADRs | 18 Core + 58 Sekolah + 24 Koperasi
15 Sprint Phases | ~100+ Database Tables

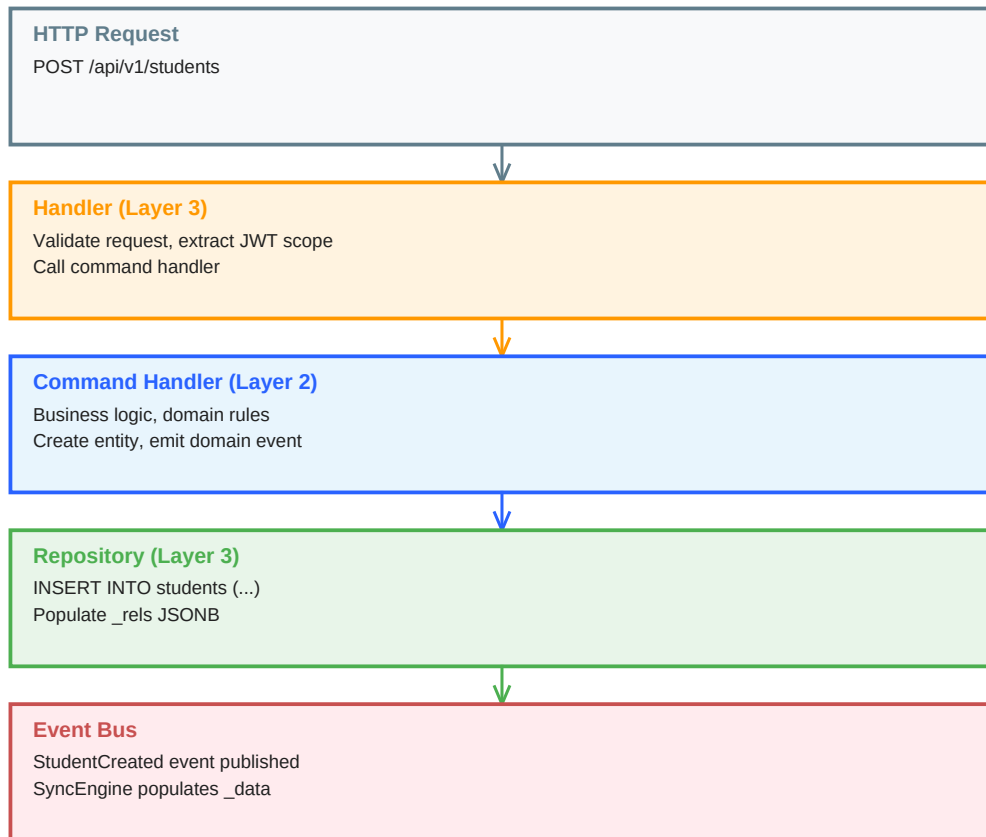
Vernon Corp Engineering | April 2026

1. High-Level System Architecture

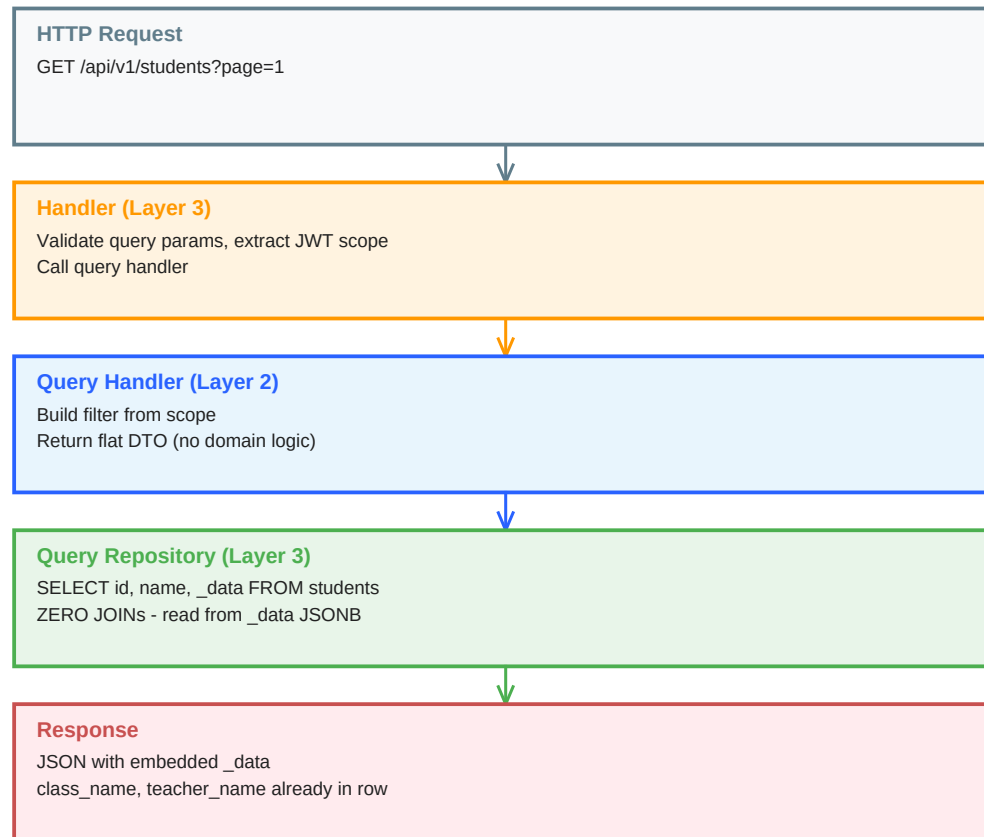


2. Clean Architecture + CQRS Flow

COMMAND (Write Path)



QUERY (Read Path)



KEY INSIGHT: Vernon Pattern eliminates JOINS on read path

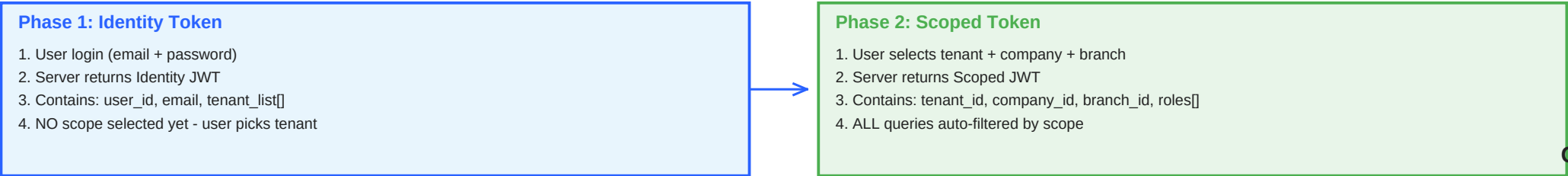
Write path: normal INSERT/UPDATE + populate _rels with FK IDs. SyncEngine (event-driven) populates _data with denormalized snapshots.
Read path: SELECT from single table, _data JSONB already contains related entity names/codes. Result: <100ms for complex listing queries.

3. Multi-Tenant & Authentication Flow

A. Four-Level Scope Hierarchy



B. Two-Phase JWT Authentication



C. Dua

school_type: "general" | "islamic"
General: SD/SMP/SMA/SMK standard
Islamic: Pondok Pesantren (Santri, Halaqah, Tahfiz)

coop_type: "general" | "islamic"
General: Koperasi konvensional (bunga)
Islamic: BMT (bagi hasil, akad syariah, zakat)

POLICY: Immutable After Activation

school_type & coop_type are set ONCE during onboarding and CANNOT be changed. Both fields are independent (4 possible combinations).

4. Vernon Denormalized Read-Cache Pattern

How `_rels` and `_data` work

TABLE: students						
id (UUID7)	name	class_id	...	_rels (JSONB)	_data (JSONB)	
abc-123	Ahmad	cls-01	...	<code>{"class_id": "cls-01", "teacher_id": "tch-05", "academic_year": "ay-02"}</code>	<code>{"class": {"name": "7A", "level": "VII"}, "teacher": {"name": "Pak Budi"}, "academic_year": {"name": "2025/2026"}}</code>	

Sync E



Benefits

BENEFITS

- + Zero JOINS on read path (<100ms)
- + Schema-flexible (new fields = JSONB update, no ALTER)
- + Read:Write asymmetry optimized (10:1 to 100:1)
- + Single-table queries = simple pagination & filtering
- + Consistent denormalization via SyncEngine

TRADE-OFFS

- Write path slightly slower (populate `_rels`)
- Storage overhead (JSONB per row)
- Eventual consistency (`_data` lags by event latency)
- SyncEngine must handle cascading updates
- Need GIN indexes on `_rels` and `_data`

5. System Policies & Business Rules

Multi-Tenant Data Isolation

- Every table has tenant_id - enforced at middleware level
- Row-Level Security: queries auto-filtered by JWT scope (tenant/company/branch)
- Cross-tenant access only for superadmin with explicit scope override
- Operational tables have full 4-level hierarchy (tenant > company > branch > warehouse)

RBAC & Authorization

- Role-based access control with hierarchical permissions
- Roles: SuperAdmin, Admin Yayasan, Kepala Sekolah, Guru, TU, Bendahara, Orang Tua, Siswa
- Koperasi roles: Manager, Teller, Nasabah, Auditor
- Permission inheritance: Yayasan roles can access all schools under their tenant

Financial Transaction Policies (Koperasi)

- Double-entry accounting (every transaction = debit + credit journal)
- Transaction immutability - no UPDATE/DELETE, only reversal transactions
- Teller session: cash must balance at session close (open -> transact -> close -> reconcile)
- Multi-level approval for loans (credit scoring + committee approval)
- NPL classification: Lancar > DPK > Kurang Lancar > Diragukan > Macet
- Regulatory compliance: OJK/LKM reporting, UU PDP (data privacy), tax integration

Academic Policies (Sekolah)

- Academic Year as universal time scope - all records tied to tahun_ajaran_id
- Student Class Placement: one active placement per student per semester
- Attendance: daily + per-subject tracking with status (Hadir/Sakit/Izin/Alpha)
- Rapor generation: orchestrates grades + attendance + extracurricular + discipline
- Dapodik integration: sync student/teacher data to national education database

5. System Policies & Business Rules (cont.)

Dual-Mode Business Rules

- Conventional: interest-based (bunga), standard koperasi terms
- Islamic (BMT): profit-sharing (bagi hasil/nisbah), syariah contracts (akad)
- BMT-specific: Zakat, Infaq, Ta'zir (social fund penalty), Rahn (collateral)
- Label mapping: terminology auto-switches based on institution type config
- Type is IMMUTABLE after tenant activation - prevents data inconsistency

Event-Driven Architecture

- Domain events published via Event Bus (InMemory for dev, NATS JetStream for prod)
- Vernon SyncEngine listens for entity changes and updates _data JSONB
- At-least-once delivery guarantee with idempotent handlers
- Dead-letter queue for failed event processing with retry + alerting

6. Domain Map - All Modules

MANAGEMENT SEKOLAH - 58 ADR across 13 domain groups

Siswa	S001-S018	Data siswa, wali, akademik, kesehatan, keuangan, disiplin, ekskul, PPDB, BK, rapor
Kurikulum	S019-S025	Kurikulum, mata pelajaran, jadwal, ujian, kalender, RPP, jurnal mengajar
Guru & HR	S026-S032	Absensi guru, beban mengajar, PKG, PKB, cuti, payroll, piket
Asrama	S033-S035	Manajemen kamar, aktivitas & absensi asrama, disiplin & kesehatan
Kantin	S036-S037	Menu & vendor management, transaksi & billing kantin
Fasilitas	S038-S041	Perpustakaan, laboratorium, aset & inventaris, booking ruangan
Komunikasi	S042-S045	Portal orang tua, messaging, notifikasi multi-channel, pengumuman
Administrasi	S046-S049	Surat menyurat, approval workflow, profil sekolah, komite
Keuangan	S050-S051	RKAS (anggaran), payment gateway
Transport	S052	Manajemen antar-jemput siswa
Integrasi	S053-S055	E-learning, reporting & analytics, Dapodik sync
Alumni	S056-S058	Alumni, beasiswa, kegiatan sekolah

MANAGEMENT KOPERASI SEKOLAH - 24 ADR across 7 domain groups

Core	K001-K003	Nasabah/anggota, rekening, produk & akad (konvensional + syariah)
Simpanan	K004-K006	Simpanan pokok/wajib, tabungan multi-produk, deposito/berjangka
Pembiayaan	K007-K010	Pinjaman/pembiayaan, angsuran, denda/penalti, jaminan/agunan
Transaksi	K011-K014	Transaction engine, teller session, denominasi uang, kas & cash flow
Akuntansi	K015-K018	Jurnal & COA, SHU, laporan regulasi (OJK/Dinas), zakat & infaq (BMT)
Extensions	K019-K021	Toko & kantin POS, payroll auto-deduction, e-wallet/uang saku digital
Platform	K022-K024	Notifikasi, dashboard & portal, integrasi & API

CORE (Shared Foundation) - 18 ADR

Architecture	ADR-001..009	Clean Arch + CQRS, Vernon Pattern, Event Bus, DI, UUID v7, Multi-Tenant, Dual-Mode
Master Data	ADR-010..013	Tahun Ajaran, Kelas & Rombel, Guru & Staff, Users & Roles (RBAC)
Infrastructure	ADR-014..018	Sync Engine, Frontend Multi-App, CI/CD, Data Migration, Regulatory Compliance
Business	BIZ-001	Business Model, Pricing Strategy, Unit Economics

7. Sprint Roadmap - 15 Phases

Sprint	Phase	Focus	ADRs	Key Deliverables
Sprint 0	Fase 0	Platform Foundation	8 ADR	Clean Arch, CQRS, Vernon, Event Bus, DI, UUID v7, React, Dual-Mode
Sprint 1	Fase 1	Auth & Multi-Tenant	2 ADR	4-Level Hierarchy, Two-Phase JWT, RBAC, User CRUD
Sprint 2	Fase 2+3	Master Data + Core Koperasi	6 ADR	Tahun Ajaran, Guru, Kelas + Produk, Nasabah, Rekening
Sprint 3	Fase 4A+4B	Student Core + Simpanan	10 ADR	Data Siswa, Wali, PPDB + Simpanan Pokok/Wajib, Tabungan, Deposito
Sprint 4	Fase 5A+5B	Kurikulum + Pembiayaan	10 ADR	Kurikulum, Jadwal, RPP + Pinjaman, Angsuran, Denda, Jaminan
Sprint 5	Fase 6A+6B	Kehadiran & Nilai + Transaksi	12 ADR	Absensi, Nilai, Ujian, Ekskul + Transaction Engine, Teller, Kas
Sprint 6	Fase 7A+7B	Rapor & HR + Akuntansi	11 ADR	Rapor, SPP, HR Guru + Jurnal COA, SHU, Laporan OJK, Zakat
Sprint 7	Fase 8	Fasilitas & Sarana	7 ADR	Perpustakaan, Lab, Aset, Booking, PKG, PKB, Payroll
Sprint 8	Fase 9+10	Asrama/Kantin + Ext.	8 ADR	Dormitory, Canteen + Toko POS, E-Wallet, Auto-Deduction
Sprint 9	Fase 11	Komunikasi & Portal	7 ADR	Notifikasi, Messaging, Portal Orang Tua, Dashboard
Sprint 10	Fase 12	Administrasi & Keuangan	6 ADR	Surat TU, Approval, Profil Sekolah, RKAS, Payment Gateway
Sprint 11	Fase 13	Integrasi & Pelaporan	5 ADR	Analytics, Dapodik, E-Learning, Transport, API Koperasi
Sprint 12+	Fase 14	Nice-to-Have	3 ADR	Alumni, Beasiswa, Kegiatan Sekolah
Ongoing	Fase INF	Infrastructure	6 ADR	Sync Engine, Frontend Arch, CI/CD, Migration, Compliance, Pricing

TOTAL: 100 ADRs | Core: 18 + Sekolah: 58 + Koperasi: 24 | ~100+ Database Tables | 16 Stakeholder Roles

8. Stakeholder Coverage

Kepala Sekolah Strategic oversight, PKG approval, analytics dashboard, accreditation	Wakasek Kurikulum Curriculum, subject management, schedule approval, RPP review
Wakasek Kesiswaan Student discipline, extracurricular, counseling, events	Wakasek Sarana Library, laboratory, asset inventory, facility booking
Guru / Teacher Attendance input, grade input, lesson plan, teaching journal, workload	Tata Usaha (TU) Correspondence, student documents, school profile
Bendahara SPP collection, RKAS budget, payment gateway, payroll	Orang Tua / Wali Parent portal, notifications, messaging, student progress
Siswa / Student Dashboard, academic records, e-wallet, attendance view	Manager Koperasi All koperasi operations, loan approval, SHU, reporting
Teller Koperasi Transaction processing, cash management, daily reconciliation	Nasabah Koperasi Self-service portal, savings, loan status, statements
Dinas Pendidikan Dapodik integration, school reporting, regulatory compliance	OJK / Regulator Financial reports, NPL ratios, CAR, compliance reports
Pustakawan/Laboran Library/lab management, booking, inventory	Komite Sekolah Budget approval, school oversight, committee management